Flint Q8 Benchmark

Jaime Lawson | Founder & Research Lead, **Phoenix Applied Technologies**

Public benchmark contact: jmlawson@outlook.com

Local AI inference, hardware optimization, offline AI infrastructure and practical model evaluation.

Release package v5 - November 5, 2026. Public release approved; not yet published.

System and model

Debian 13; AMD Ryzen 7 2700X; approximately 78 GiB RAM; RTX 4060 8 GB + GTX 1080 Ti 11 GB. Flint Q8_0: **42.30 GB**; Colibri v2.1.0. Later CUDA tests used eight inference CPU workers, dense computation on the 4060, and expert caches on both GPUs. These are whole-system results, including CPU/RAM, not standalone GPU scores.

Results at a glance

All rates include reasoning. Completed rates use final token counts / decode-loop time; sampled rates are the last active cumulative average.

Test	Decode rate	End-to-end time	Evidence status
Full explanatory response, control A	30.464 tok/s	61.15-61.94 s	Completed; 1,819 tokens
Full explanatory response, control B	**30.567 tok/s**	60.14-60.94 s	Completed; same 1,819-token output
Full coding repair	**20.418 tok/s**	**242.71-243.50 s**	Completed; fix passed 7/7 tests
768-token explanatory screen	**31.194 tok/s**	25.24-26.03 s	**Preliminary** sampled rate; truncated

The two explanatory controls verified similar performance on the same machine and prompt, with identical exported final text and reasoning. They are not independent replications. **31.2 tok/s remains preliminary:** one capped run, no same-protocol independent repeat, and an unresolved early UI-temperature caveat. Sources: E1-E4 in the technical report.

Optimization and correctness

Automatic CUDA cache allocation measured 31.194 versus 27.830 tok/s with a fixed allowance in the short screen, a **12.1% observed increase**, subject to the same preliminary caveat. On later full responses, prompt-aware cache replanning was **14.3% slower**, and moving dense computation to the Ti was **11.0% slower**. These are workload-specific observations; faster decode did not always mean a shorter answer.

The async-cache repair passed seven predefined checks in both saved completed attempts. The first single-GPU attempt had truncated before a final answer. The baseline code fails two tests. **The physics failure remains documented:** several insulation responses misrepresented GPU thermal-interface materials as barriers that improve cooling by slowing heat toward the cooler. The explanatory timing test was not an accuracy pass. No general coding or accuracy score is claimed.

Evidence and credits

The four-page technical report, raw evidence and verified offline test commands accompany this summary. Credit: Colibri developers, including Vincenzo Fornaro; Recursive for Flint; Unsloth for GGUF packaging. Codex/Sapphire assisted preparation and verification; this is not independent replication. See CREDITS.md and LICENSING.md for attribution and license scope.

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